

PREPARED FOR

T6N, R4W, Mount Diablo Meridian

COORDINATES 38.3459°N 122.3047°W WGS84	ELEVATION 66 ft NAVD88 · MSL	STATE PLANE CA 2 (0402) · N 1,887,922.81 E 6,474,289.05 (US ft, NAD83)	FEMA ZONE Zone X 06055C0510FF	DECLINATION 13.0° E WMM 2026
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APN 038050009000 | ZONING AP | ACRES 0.481 ac

BEST GNSS WINDOW Mon, Jul 13 · 6:00 AM – 10:00 AM | PDOP 0.99 · 34 sv | NEAREST CORS
P261 · 14.1 mi ✓

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Site file · Google Earth
Download KMZ

RECORDER
Napa County Recorder-Clerk · 1127 First
St, Napa, CA 94559

NEAREST CORS
P261 · 14.1 mi

GENERATED
July 13, 2026

ASSESSOR
Napa County Assessor · 1127 First St,
Napa, CA 94559

NEAREST BM
JT9010 · 0.2 mi

SYSTEM
TopoScout · info@toposcout.com



This report compiles data from public and third-party sources (NGS, FEMA, USGS, NOAA, and county records) current as of the generation date shown above. GNSS planning values are predictive and conditions may change. This document is provided for planning purposes only and does not constitute a boundary survey, legal land description, or engineering certification. Verify all critical values against primary sources.

SITE & SIGNAL REPORT

COORDINATES

38.3459, -122.3047

WGS84 / NAD83

ELEVATION

66 ft

Above MSL (approx)

MAGNETIC DECLINATION

13.0° E

Compass reads 13.0° too far W

FEMA FLOOD ZONE

X

AREA OF MINIMAL FLOOD HAZARD · Panel 06055C0510FF

IONOSPHERIC CONDITIONS

PEAK KP INDEX

4.7

AVG KP

2.4

SOLAR FLUX F10.7

138 sfu

TEC (VTEC) ▲ DEGRADED

17.6 TECU

-5.5 TECU vs median

ASSESSMENT

MODERATE ACTIVITY

Peak Kp 4.7 (avg 2.4). Solar flux F10.7 = 138 sfu. Moderate geomagnetic activity. Some ionospheric delay possible. Static sessions should use longer occupation times.

GNSS / DOP SUMMARY

DATE	AVG PDOP	BEST 4-HR WINDOW	AVG SATS	RATING
Mon, Jul 13	0.99	6:00 AM – 10:00 AM	34	Ideal

GPS + GLONASS + Galileo + BeiDou MEO · Full multi-constellation model · 15° elevation mask · Per-day independent computation
GNSS satellite geometry is accurate within a 50-mile radius of the entered location – orbital mechanics do not change meaningfully at ground-level distances. Weather data is specific to the entered coordinates.



WILDFIRE PROXIMITY

No Active Fires Within 150 Miles

No active fires as of report date (July 13, 2026) · Conditions may change – verify before fieldwork



GPS CONSTELLATION HEALTH

▲ 1 Satellite Currently Unhealthy – PRN 11

Active outage per NAVCEN · Affected satellites auto-excluded by receiver

Land Cover & Multipath Risk

SKY OBSTRUCTION

MULTIPATH RISK

Minor Obstruction

Low vegetation or sparse development nearby — minor sky obstruction possible.

Low Multipath

Minor reflective surfaces present — multipath generally manageable.

RECOMMENDED ELEVATION MASK

13° Minor vegetation — slight horizon reduction

✓ PDOP Check at 13°: 0.8–0.8 across survey window
PDOP remains ≤ 0.8 at 13° — geometry confirmed acceptable

5-POINT SAMPLE (CENTER + N/S/E/W · ~100 M)

Point	Cover Type	Obstruction	Multipath
Center	Dwarf Scrub	Low	Low
North	Dwarf Scrub	Low	Low
South	Dwarf Scrub	Low	Low
East	Dwarf Scrub	Low	Low
West	Dwarf Scrub	Low	Low

Dominant cover: Dwarf Scrub · Source: USGS NLCD 2024 via MRLC · 30 m resolution · Industry standard mask guidance per NGS field procedures

Monday, Jul 13

Overcast

9 mph

GO

High 93°F

Low 59°F

Wind max 9 mph

Gusts 12 mph

Precip 0.00"

Sunrise 5:56 AM

Sunset 8:33 PM

Waning Crescent 2%

GNSS: Avg PDOP **0.99** · Best 4-hr window 6:00 AM – 10:00 AM (PDOP 0.94, 34 sats avg)

GNSS / DOP — HOURLY

	PDOP	SATS	RATING
6:00 AM	0.84	36 sv	IDEAL
7:00 AM	0.95	34 sv	IDEAL
8:00 AM	0.95	33 sv	IDEAL
9:00 AM	1.00	33 sv	GOOD
10:00 AM	1.01	29 sv	GOOD
11:00 AM	1.11	25 sv	GOOD
12:00 PM	1.03	26 sv	GOOD
1:00 PM	1.08	26 sv	GOOD
2:00 PM	0.93	32 sv	IDEAL
3:00 PM	0.97	33 sv	IDEAL
4:00 PM	0.98	33 sv	IDEAL
5:00 PM	0.99	33 sv	IDEAL
6:00 PM	1.02	30 sv	GOOD
7:00 PM	0.99	32 sv	IDEAL



GO — GOOD CONDITIONS

Partly cloudy | Wind: 9.4 mph (gusts 11.6 mph) | Precip: 0" | Temp: 59°–93.2°F



DRONE WIND ANALYSIS



GO Peak: 9.4/11.6 mph S

* Sunrise: 5:56 AM Sunset: 8:33 PM fly window clipped to daylight

Best flight window: 6:00 AM – 9:00 PM (15 hrs)

Wind direction shifting — plan mission headings carefully

SUSTAINED / GUSTS mph ≤turbulence

	SUSTAINED / GUSTS mph	≤turbulence	DIR
5 AM	2/4 mph		SE
6 AM	0/3 mph		W
7 AM	3/5 mph		E
8 AM	2/2 mph		SSW
9 AM	2/3 mph		SE
10 AM	4/5 mph		SSE
11 AM	5/5 mph		S
12 PM	5/6 mph		S
1 PM	5/5 mph		S
2 PM	9/10 mph		S
3 PM	7/8 mph		S
4 PM	8/9 mph		SSW
5 PM	9/11 mph		SSW
6 PM	9/11 mph		SSW
7 PM	9/12 mph		S
8 PM	8/11 mph		S

BEST WINDOW

6:00 AM – 9:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE CLASS A ► FREE TO FLY

Class G uncontrolled airspace — fly under 400ft AGL, follow Part 107

→ TFR STATUS ☒ NO TFRs WITHIN 30 MI [Check B4UFLY](#)

Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

FIELD PLANNING REPORT

GNSS SURVEY OPERATIONS — NAPA COUNTY

Generated: Monday, July 13, 2026

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VERDICT: GO

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Primary Reason: Excellent PDOP geometry (avg 0.99, peak window 0.94 with 34 satellites), benign weather (overcast, light winds, no precipitation), and moderate ionospheric conditions manageable through standard static occupation protocols.

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BEST SURVEY WINDOW

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0600–1000 hours (4 hours)

Why: PDOP 0.94 at peak (best in 24-hour window), 34-satellite constellation (GPS+GLONASS+Galileo+BeiDou), 15° elevation mask. Morning session avoids afternoon thermal effects and maximizes geometric strength. Overcast conditions reduce multipath risk from ground reflections.

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BEST 4-HOUR STATIC GNSS SESSION

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0630–1030 hours

Configuration: 30-second data interval (standard); 15° elevation mask. Given moderate geomagnetic activity (Kp avg 2.4, peak 4.7) and TEC slightly below normal (−5.5 TECU), occupy full 4 hours to mitigate ionospheric delay bias. Multi-constellation receivers (GPS+GLONASS+Galileo+BeiDou) strongly recommended. Position convergence expected within 20–30 minutes; use full session for optimal ambiguity resolution and error characterization.

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DRONE OPERATIONS ASSESSMENT

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ACCEPTABLE. Overcast conditions provide diffuse light and stable thermal conditions. Wind maxima 9 mph (manageable). No wildfire smoke detected. Standard airspace coordination required per local NPS/county regulations. No GNSS-specific constraints on UAS operations this session.

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FIELD SAFETY WARNINGS

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MODERATE GEOMAGNETIC ACTIVITY: Kp 4.7 peak may cause minor ionospheric disturbance. Receivers with weak signal handling may show brief tracking losses. Multipath rejection filters recommended. No field-crew hazard.

SATELLITE HEALTH: PRN 11 currently unhealthy (UNUSUFN); receivers will auto-exclude. PRN 3 and PRN 26 scheduled maintenance windows after survey date — informational only. Constellation strength remains excellent.

NO WILDFIRE THREAT: No active fires within detection range. Air quality nominal.

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SUMMARY

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Conditions strongly favor commencement on 13 July 2026. Multi-constellation GNSS geometry is excellent (PDOP 0.94 in optimal morning window 0600–1000 hrs), weather is stable and clear of precipitation, and ionospheric activity is moderate and manageable with standard 4-hour static protocols. Occupy primary session 0630–1030 to maximize ambiguity resolution under current geomagnetic moderate-activity regime. Site terrain (dwarf scrub, low obstruction) and satellite constellation health support high-confidence positioning. Proceed with survey as planned.

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COUNTY RECORDING OFFICES

RECORDER / REGISTER OF DEEDS

Napa County Recorder-Clerk

1127 First St, Napa, CA 94559

ASSESSOR / TAX OFFICE

Napa County Assessor

1127 First St, Napa, CA 94559

CORS STATIONS – WITHIN 30 MILES

QR / LINK	STATION ID	NAME / LOCATION	DISTANCE
 NGS ↗	P261	HUNTERHILLCN2004	14.1 mi
 NGS ↗	P198	PETALUMAIRC2004	17.5 mi
 NGS ↗	OHLN	OHLONE PARK	23.5 mi
 NGS ↗	P196	MEACHUMLFLCN2006	24.0 mi
 NGS ↗	CASR	SANTA ROSA CA	24.9 mi
 NGS ↗	P267	DIXONAVIATCN2005	26.3 mi

NGS BENCHMARKS – WITHIN 10 MILES

QR / LINK	PID	DESIGNATION	ELEVATION	ORDER	DISTANCE
 Datasheet ↗	JT9010	"13"	19.100m / 62.7ft NAVD88	Order	0.2 mi
 Datasheet ↗	JT9018	"13 E"	17.970m / 59.0ft NAVD88	Order	0.2 mi
 Datasheet ↗	JT9009	"13"	17.900m / 58.7ft NAVD88	Order	0.2 mi
 Datasheet ↗	JT9017	"13 D"	17.890m / 58.7ft NAVD88	Order	0.2 mi
 Datasheet ↗	JT9034	"20 A"	20.730m / 68.0ft NAVD88	Order	0.3 mi
 Datasheet ↗	JT9012	"13"	19.060m / 62.5ft NAVD88	Order	0.4 mi
 Datasheet ↗	JT9043	"25 A"	18.710m / 61.4ft NAVD88	Order	0.4 mi
 Datasheet ↗	JT9016	"13 D"	18.200m / 59.7ft NAVD88	Order	0.5 mi

QR / LINK	PID	DESIGNATION	ELEVATION	ORDER	DISTANCE
Datasheet ↗					

FEMA FLOOD ZONE

X – AREA OF MINIMAL FLOOD HAZARD

Panel: 06055C0510FF

Effective: Sep 29, 2010

[View FEMA Map ↗](#)



DATA COLLECTOR IMPORT – NGS BENCHMARKS

↓ Download .txt File

NGS BENCHMARK CONTROL POINTS – Copy / Paste to Data Collector

Datum: NAD83 | Vertical: NAVD88 | Source: NGS NDE API

PID	NAME	LAT	LON	ELEV_FT	ORDER
JT9010	13	38.348333	-122.302778	62.7	
JT9018	13 E	38.348333	-122.302778	59.0	
JT9009	13	38.346667	-122.301111	58.7	
JT9017	13 D	38.345000	-122.301111	58.7	
JT9034	20 A	38.345000	-122.309444	68.0	
JT9012	13	38.351667	-122.304444	62.5	
JT9043	25 A	38.340000	-122.302778	61.4	
JT9016	13 D	38.341667	-122.297778	59.7	

USGS STREAM GAUGES – within 25 miles

STATION	DIST	LEVEL	FLOW	FLOOD STATUS
NAPA R NR NAPA CA #11458000	1.5 mi	2.37 ft	4.39 cfs	● Normal
NATHANSON C A SONOMA CA #11458600	9.3 mi	—	—	● Normal
SONOMA C A AGUA CALIENTE CA #11458500	10.4 mi	8.84 ft	2.73 cfs	● Normal
NAPA R NR ST HELENA CA #11456000	14.1 mi	1.43 ft	3.23 cfs	● Normal
SONOMA CREEK A KENWOOD CA #11458433	14.9 mi	3.93 ft	0.19 cfs	● Normal
PUTAH C NR WINTERS CA #11454000	16.8 mi	7.85 ft	578 cfs	— Unknown
SUISUN SLOUGH NR BENICIA CA #11455740	19.6 mi	13.03 ft	2,230 cfs	— Unknown
POPE C A WALTER SPRINGS CA #11453590	21.1 mi	—	—	— Unknown

USGS LAKES & RESERVOIRS – within 150 miles

STATION	DIST	LEVEL	FULL POOL	LINK
CLEAR LK A LAKEPORT CA #11450000	58.1 mi	5.03 ft	N/R	USGS ↗

CALIFORNIA RESERVOIRS (DWR/CDEC) – within 150 miles

RESERVOIR	DIST	ELEVATION	FULL POOL	STORAGE	% CAP
Lake Del Valle	60.6 mi	696.94 ft (-9 ft)	706 ft	36K af	67%
Folsom Lake	66.5 mi	448.16 ft (-18 ft)	466 ft	787K af	81%
Lake Oroville	93.4 mi	861.14 ft (-39 ft)	900 ft	2,858K af	81%
New Melones Lake	100.1 mi	1,020.15 ft (-68 ft)	1,088 ft	1,664K af	69%
Black Butte Lake	101.2 mi	—	228 ft	74K af	54%

USBR RESERVOIRS (Bureau of Reclamation) – within 150 miles

RESERVOIR	DIST	ELEVATION	FULL POOL	STORAGE	% CAP
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ARMY CORPS RESERVOIRS (USACE) – within 150 miles

RESERVOIR	DIST	POOL ELEV	FULL POOL	STORAGE	% CAP
Del Valle	64 mi	696.9 ft	695.01 ft	—	100%
Camanche	89 mi	230.3 ft	221.86 ft	—	104%
Englebright Lake-Pool	95 mi	525.1 ft	522.35 ft	—	101%
Farmington Dam-Pool	99 mi	124.9 ft	117.09 ft	—	107%
Black Butte-Pool	101 mi	457 ft	458.36 ft	—	100%
Don Pedro	137 mi	—	N/R	—	—

USGS GROUNDWATER WELLS – within 15 miles

STATION	DIST	WELL DEPTH	WATER LEVEL	DATE	AQUIFER
005N004W32A002M	7 mi	200 ft	87.71 ft BGS	2014-11-05	N100CACSTL
005N004W28R001M	7 mi	51 ft	41.90 ft BGS	1977-10-03	N100CACSTL
004N004W04C001M	8 mi	80 ft	11.20 ft BGS	1977-10-03	N100CACSTL
004N004W05B001M	8 mi	—	54.10 ft BGS	1977-10-03	N100CACSTL
005N004W31R001M	8.1 mi	295 ft	—	—	—
004N004W05D002M	8.3 mi	60 ft	14.60 ft BGS	1977-10-03	N100CACSTL

004N004W06B001M	8.3 mi	201 ft	21.86 ft BGS	2022-06-07	—
004N004W02L001M	8.8 mi	—	9.90 ft BGS	1977-10-02	N100CACSTL

🔍 AI AQUIFER INTELLIGENCE — CLAUDE HAIKU

The Coast Ranges Sedimentary aquifer (N100CACSTL) is composed of layered sand, gravel, and clay deposits formed during marine regression and alluvial infilling in the Tertiary and Quaternary periods. This mixed lithology creates variable water-bearing zones separated by confining layers, typical of California's Coast Range basins. Water availability at this site depends on striking productive sand/gravel lenses; the wide spread in well depths (51–295 ft) and water levels (9.90–87.71 ft) indicates heterogeneous storage and transmission—some zones yield readily, others do not.

FORMATION

Name	Coast Ranges Sedimentary (N100CACSTL)
Age / Rock	Tertiary to Quaternary sand, gravel, silt, and clay deposited in a subsiding marine-to-fluvial basin.
Extent	800–1200 sq mi (est. USGS/DWR)

DEPTH & RECHARGE

Depth Range	51–295 ft (5–30 floors)
Typical Depth	100–150 ft (10–15 floors)
Recharge Source	Precipitation on surrounding uplands and hillslope convergence; limited lateral inflow from higher-elevation metamorphic terranes.
Transit Time	15–50 years to 100 ft depth (est. USGS/DWR)

WATER QUALITY

pH	est. 6.9–7.4 (est. USGS/DWR)
Hardness	est. 180–350 mg/L CaCO3 (est. USGS/DWR)
TDS	est. 280–500 mg/L (est. USGS/DWR)
Notes	No sample data — typical for formation. Regional water generally suitable for domestic use; monitor for seasonal salinity in proximity to marine transgressions.

TREND

Trend	Mixed: shallow wells show recovery from 1977 lows (9.90–14.60 ft) to 2022 levels (21.86 ft); deeper well shows decline from 1977 (54.10 ft) to 2014 (87.71 ft), suggesting continued abstraction stress or localized confinement effects.
Source	USGS NWIS historical record and site measurements.
Seasonal Swing	est. 8–20 ft (est. USGS/DWR); expect greatest drawdown in late summer/fall.

Source: USGS National Water Information System · Search all wells in this area → USGS NWIS Mapper ↗ · Water level data pending USGS API migration for CA state well numbers

LAND SUBSIDENCE & AQUIFER COMPACTION

✓ LOW SUBSIDENCE RISK — No known critically overdrafted basin

Source: CA DWR SGMA Bulletin 118 2024.1 · California only - full US pending server migration

🔍 AI SUBSIDENCE ASSESSMENT — CLAUDE HAIKU

The Napa-Sonoma region (aquifer N100CACSTL) is underlain by well-drained soils and consolidated bedrock typical of coastal California uplands, where groundwater extraction has not historically induced significant aquifer-system compaction. The site lies outside designated SGMA critically overdrafted basins, indicating that regional groundwater management has maintained sustainable withdrawal rates without triggering the clay-layer consolidation that causes subsidence in overexploited basins. For surveyors, this means historical and recent elevation control should remain reliable without the need for subsidence corrections.

SUBSIDENCE ASSESSMENT

Susceptibility	Low
Primary Mechanism	None - consolidated terrain
Annual Rate	<1 mm/yr (est. USGS)
Historic Loss	None documented
Aquifer Compaction	None - aquifer not over-pumped
SGMA Status	Not applicable - not in SGMA basin

BENCHMARK IMPLICATIONS

Monument Risk	Low - monuments stable
Elevation Validity	Good - no active subsidence
GNSS Recommendation	Standard benchmark verification adequate

SURVEY ACTION

Required Action	No special action required
Dataset	CA DWR SGMA Bulletin 118 2024.1
Coverage	California critically overdrafted basins only - full US coverage pending

STRUCTURAL & ENGINEERING PARAMETERS

SEISMIC (ASCE 7-22)

Site Class	D (default)
Risk Category	II

WIND / SNOW / ICE / FROST (ASCE 7-22)

Basic Wind Speed	92 mph
Ground Snow Load	5.8 psf
Winter Wind (W2)	0.35
Ice Thickness	None
Frost Depth	12 in minimum

Est. from IECC Zone 3C — AFI data unavailable

CLIMATE ZONE (IECC 2021): Zone 3C · Marine · Marine - mild year-round, coastal influence (SF, LA coast) County: Napa

AI CLIMATE SUMMARY — RESIDENTIAL DESIGN

This marine climate stays mild year-round with minimal temperature swings, so your primary envelope concern is managing moisture and wind rather than extreme heat or cold—prioritize air sealing and durable cladding that can handle salt spray and wind-driven rain. Maximize south and west-facing glazing to capture solar gain during cooler months while using deep overhangs to shade summer sun, but minimize north-facing glass and keep windows away from the prevailing wind direction to reduce infiltration and heat loss. Your best passive design move is cross-ventilation: design for strong natural ventilation pathways that take advantage of persistent coastal breezes, which will let you reduce or eliminate mechanical cooling on the frequent mild days when residents can simply open windows.

AI STRUCTURAL ASSESSMENT — CLAUDE HAIKU

This coastal California site presents moderate wind exposure and modest snow loading with well-drained soils suitable for conventional foundations. Seismic parameters are currently unavailable and may warrant confirmation with the local building official before final design; consult a licensed engineer to establish appropriate seismic detailing requirements. Frost depth of approximately 12 inches suggests standard foundation design practices for the region.

FOUNDATION

Type Conventional spread footing anticipated

Basis Well-drained gravelly loam with moderate slopes suggests adequate bearing capacity; however, seismic site class D and unavailable SDC parameters may warrant geotechnical investigation to confirm design criteria—consult engineer.

STRUCTURAL IMPLICATIONS

Seismic Seismic design category unavailable—recommend consulting local building official and licensed engineer to determine applicable detailing; Site Class D suggests potential for elevated seismic response that may warrant special moment frames or intermediate systems.

Wind Exposure Exposure C (open terrain) anticipated per ASCE 7-22 Risk Category II

Wind 92 mph 3-second gust wind speed anticipated; standard wind load design likely adequate—no coastal hurricane exposure anticipated for this inland location.

Snow Ground snow load of 5.8 psf anticipated—standard roof design likely adequate; winter wind factor W2=0.35 suggests modest wind-on-snow considerations.

INVESTIGATIONS & INSPECTIONS

Geotech Required Yes—geotechnical investigation recommended to confirm Site Class D, establish bearing capacity, and verify seismic parameters for design.

Liquefaction Screen Screening recommended—confirm groundwater depth and soil saturation; Site Class D + well-drained soils suggest low to moderate liquefaction risk, but verification recommended.

Special Inspections May be required depending on final seismic design category determination—consult engineer and local building official.

Code Edition ASCE 7-22 / IBC 2021 (planning reference only)

Site Class Note Site Class D assumed for planning—geotechnical investigation recommended to confirm soil profile, groundwater conditions, and seismic site class classification prior to final design.

SOILS — USDA SOIL SURVEY

MAP UNIT: Coombs gravelly loam, 2 to 5 percent slopes

SERIES: Coombs

DRAINAGE CLASS: Well drained

HYDRO GROUP: C

SOIL ORDER: Alfisols

SUBORDER: Xeralfs

AI SOIL INTELLIGENCE — CLAUDE HAIKU

Coombs gravelly loam is a well-drained Alfisol (Xeralf suborder) formed in alluvial or colluvial material derived from mixed sources in the Coast Ranges and inland valleys of California. The soil profile features a gravelly loam surface layer overlying a clay-enriched substratum, typical of Mediterranean-climate upland soils. Surveyors and developers should expect moderate gravel content affecting excavation and compaction, generally good bearing capacity for standard foundation work, and minimal drainage concerns on these gentle slopes.

Mixed alluvial/colluvial parent material; upland slopes and foothill positions; oak-grassland and chaparral vegetation.

DRAINAGE

Class Well drained

Seasonal Saturation None within 60 inches

Flooding Risk Minimal on 2–5% slopes

Ponding Risk None

ENGINEERING

Bearing Capacity 2,500–3,500 psf (shallow foundations)

Rating Good

Shrink-Swell Moderate — clay-enriched substratum typical of Xeralfs

Frost Action Low

Cut Suitability Good (gravelly material is stable and drains well)

Fill Suitability Fair (subsoil clay content requires moisture control during compaction)

SEPTIC

Suitability Conditional

Notes Perc test required — clay substratum may limit percolation rate; depth to restrictive layer should be verified

HAZARDS

Excavation Moderate — 20–35% gravel content may require equipment adjustment

Expansive Clay Low to moderate — monitor clay substratum during wet season

Erosion Low to moderate on gentle slopes; higher risk if vegetation removed or drainage altered

Data sources: USGS National Water Information System · [waterdata.usgs.gov](#) · California DWR / CDEC · [cdec.water.ca.gov](#) · USBR RISE · [data.usbr.gov](#) · USACE CWMS · [water.usace.army.mil](#) · USGS NWIS Groundwater · [waterservices.usgs.gov](#) · USDA Web Soil Survey · [sdmdataaccess.nrcs.usda.gov](#) · CA DWR SGMA Bulletin 118 · [gis.water.ca.gov](#) · ASCE 7-22 Hazard Tool · [gis.asce.org](#) · USGS Design Maps · [earthquake.usgs.gov](#) · IECC 2021 Climate Zones · DOE Building America · Real-time data subject to revision

Water levels current as of report run: Jul 13, 2026, 3:04 AM UTC

* Live pool elevation not reported by this reservoir — full pool design elevation shown

 OIL & GAS WELLS

California — within 10mi radius

8

total

0

active

7

plugged

By status:

7

Plugged

1

Idle

By type: 8 DH

Lease / Well	Status	Type	Operator	Depth	Dist
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	7.2 mi
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	7.2 mi
McKenzie-Hazard Syndicate	Plugged	DH	McKenzie-Hazard Syndicate	—	7.4 mi
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	8.6 mi
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	8.8 mi
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	9.5 mi
Capell Oil Corp.	Idle	DH	Capell Oil Corp.	—	9.7 mi
Cordelia Oil & Gas Co.	Plugged	DH	Cordelia Oil & Gas Co.	—	12 mi

AI Assessment — Well Proximity

Well proximity is not a planning concern for this site. The nearest well is a plugged Mobil Oil well 7.2 miles away, well beyond any actionable distance threshold for foundation review, surface easements, or environmental assessment. With no active producers, injection wells, or idle wells within relevant distances, standard oil and gas well due diligence can proceed with minimal focus on subsurface legacy well impacts.

Source: California public well records. Active = currently producing or permitted. Plugged = abandoned/decommissioned. Data updated nightly.

200 **10** **108**
 total in search area within 5 mi within 25 mi

Site Name	Status	Commodities	Distance
Roderick Sand and Gravel Pit	Occurrence	Sand/Gravel	1.6 mi
Basalt Rock Co	Producer	Clay	2 mi
Cicero Pumice Quarry	Occurrence	PUM	2.9 mi
Mount George King Pumice Deposit.	Occurrence	PUM	2.9 mi
Basalt Rock Co.	Occurrence	DIT	3.1 mi
Mt George Pumice Deposit	Producer	PUM	3.1 mi
Wilson Pumice Quarry	Occurrence	PUM	3.7 mi
Unknown	Occurrence	Mercury	4.4 mi
Silva Pumice Quarry	Occurrence	PUM	4.7 mi
Silva Pumice Quarry	Producer	PUM	5 mi
Walker Pumice Quarry	Occurrence	PUM	5.1 mi
Walker Pumice Quarry	Producer	PUM	5.1 mi
Mountain	Occurrence	Mercury	5.4 mi
Unknown	Occurrence	Mercury	7.6 mi
Unknown	Occurrence	Stone/Crushed	7.8 mi
Unnamed Location	Past Producer	Sand/Gravel	8.5 mi
Unnamed Location	Past Producer	Sand/Gravel	8.5 mi
Unnamed Prospect	Occurrence	Chromium	8.8 mi
Unnamed Location	Occurrence	Chromium	9 mi
Unknown	Occurrence	Chromium	9.1 mi
Bella Oak	Past Producer	Mercury	9.4 mi
Unknown	Occurrence	MG	9.7 mi
Serres Pit	Past Producer	Sand/Gravel	10.4 mi
Unnamed Location	Unknown	Sand/Gravel	10.5 mi
Conn Valley	Occurrence	Manganese	10.6 mi

Showing 25 of 200 deposits. Full dataset via USGS MRDS.

AI Assessment — Mining Context

Three active mineral producers operate within 5 miles of the site: Basalt Rock Co (2 mi, clay operations), Mt George Pumice Deposit (3.1 mi), and Silva Pumice Quarry (5 mi). Expect heavy truck traffic on access roads during operating hours and potential dust generation from all three operations; blasting and crushing noise should be anticipated from the pumice quarries. No inactive deposits warrant Phase 1 assessment consideration at this location.

Source: USGS Mineral Resources Data System (MRDS). Status: Producer = active/recent; Past Producer = historical; Prospect = explored not mined; Occurrence = mineralization noted.

ENDANGERED & THREATENED SPECIES - USFWS Critical Habitat within 25mi

6 **6** **12**
 Endangered Threatened Total w/ Critical Habitat

6 Endangered species with designated critical habitat. ESA Section 7 consultation may be required for federal nexus projects.

COMMON NAME	SCIENTIFIC NAME	STATUS
California tiger Salamander	<i>Ambystoma californiense</i>	Endangered
Conservancy fairy shrimp	<i>Branchinecta conservatio</i>	Endangered
Contra Costa goldfields	<i>Lasthenia conjugens</i>	Endangered
Soft bird's-beak	<i>Cordylanthus mollis</i> ssp. <i>mollis</i>	Endangered
Suisun thistle	<i>Cirsium hydrophilum</i> var. <i>hydrophilum</i>	Endangered
Vernal pool tadpole shrimp	<i>Lepidurus packardii</i>	Endangered
Alameda whipsnake (=striped racer)	<i>Masticophis lateralis euryxanthus</i>	Threatened
California red-legged frog	<i>Rana draytonii</i>	Threatened
Delta smelt	<i>Hypomesus transpacificus</i>	Threatened
Northern spotted owl	<i>Strix occidentalis caurina</i>	Threatened
Vernal pool fairy shrimp	<i>Branchinecta lynchi</i>	Threatened
western snowy plover	<i>Charadrius nivosus nivosus</i>	Threatened

AI SPECIES & ESA ASSESSMENT - CLAUDE HAIKU

ESA Implications Summary – 38.3459, -122.3047 **ESA Section 7 Triggers & Planning Relevance:** Federal nexus (permits, funding, or agency involvement) triggers mandatory Section 7 consultation if your project may affect any listed species or their critical habitat. Given 12 species with designated critical habitat within 25 miles, the most planning-relevant species are **California tiger salamander, California red-legged frog, vernal pool invertebrates (fairy shrimp, tadpole shrimp), and Delta smelt**—these typically indicate wetland/vernal pool presence and hydrologic connectivity. The site's location in the San Francisco Bay Area/Central Valley ecotone suggests potential jurisdictional waters (USACE) and state water rights implications, which commonly trigger Section 7 consultation requirements. **Recommended Next Step:** Conduct a biological resources assessment including wetland delineation and focused surveys for listed species occurrence before finalizing site plans. Engage with USFWS (for terrestrial/vernal pool species) and NOAA Fisheries (for Delta smelt) early to determine if informal consultation, biological assessment, or formal Section 7 consultation is required based on project specifics. --- *For planning purposes only. Consult a licensed environmental biologist for formal ESA compliance.*

Source: USFWS Critical Habitat Final Designated Areas - FWS Open Data - Critical habitat does not equal species range

NATIVE PLANTS - iNaturalist Research-Grade within 10mi

57	5051	2026-07-04
Unique Species	Total Observations	Most Recent
COMMON NAME	SCIENTIFIC NAME	LAST OBS
American black nightshade	<i>Solanum americanum</i>	2026-05-29
Bearded Clover	<i>Trifolium barbigerum</i>	2026-05-15
bird's foot cliffbrake	<i>Pellaea mucronata</i>	2026-04-28
Blow Wives	<i>Achyrrachaena mollis</i>	2026-05-20
blue oak	<i>Quercus douglasii</i>	2026-05-09
Bolander's Phacelia	<i>Phacelia bolanderi</i>	2026-04-27
box elder	<i>Acer negundo</i>	2026-05-18
Braun's Giant Horsetail	<i>Equisetum telmateia braunii</i>	2026-06-22
California bay	<i>Umbellularia californica</i>	2026-06-24
California beeplant	<i>Scrophularia californica</i>	2026-05-20
California black oak	<i>Quercus kelloggii</i>	2026-05-15
California buckeye	<i>Aesculus californica</i>	2026-06-02
California Dutchman's Pipe	<i>Aristolochia californica</i>	2026-05-26
California milkwort	<i>Rhinotropis californica</i>	2026-05-15
California mugwort	<i>Artemisia douglasiana</i>	2026-06-02
California poppy	<i>Eschscholzia californica</i>	2026-06-24
California Wild Rose	<i>Rosa californica</i>	2026-05-09
Canyon liveforever	<i>Dudleya cymosa</i>	2026-05-30
-	<i>Castilleja densiflora densiflora</i>	2026-05-09
clay mariposa lily	<i>Calochortus argillosus</i>	2026-05-15
coast live oak	<i>Quercus agrifolia</i>	2026-05-13
coastal woodfern	<i>Dryopteris arguta</i>	2026-05-20
common selfheal	<i>Prunella vulgaris</i>	2026-06-22
Diogenes' lantern	<i>Calochortus amabilis</i>	2026-05-15
distant phacelia	<i>Phacelia distans</i>	2026-05-15
-	<i>Downingia concolor concolor</i>	2026-05-15
Dwarf Brodiaea	<i>Brodiaea terrestris</i>	2026-05-15
Elegant Clarkia	<i>Clarkia unguiculata</i>	2026-05-16

COMMON NAME	SCIENTIFIC NAME	LAST OBS
goldback fern	<i>Pentagramma triangularis</i>	2026-05-20
Ithuriel's Spear	<i>Triteleia laxa</i>	2026-06-15
maroonspot calicoflower	<i>Downingia concolor</i>	2026-05-15
narrowleaf milkweed	<i>Asclepias fascicularis</i>	2026-06-25
narrowleaf mule-ears	<i>Wyethia angustifolia</i>	2026-05-15
Narrowleaf Willow	<i>Salix exigua</i>	2026-06-24
needleleaf navarretia	<i>Navarretia intertexta</i>	2026-05-15
northern California black walnut	<i>Juglans hindsii</i>	2026-06-24
orange bush monkeyflower	<i>Diplacus aurantiacus</i>	2026-05-29
Oregon Ash	<i>Fraxinus latifolia</i>	2026-06-02
Pacific Bleeding Heart	<i>Dicentra formosa</i>	2026-05-04
Pacific madrone	<i>Arbutus menziesii</i>	2026-07-04
Pacific poison oak	<i>Toxicodendron diversilobum</i>	2026-06-02
pineapple-weed	<i>Matricaria discoidea</i>	2026-05-20
Redwood Sorrel	<i>Oxalis smalliana</i>	2026-05-28
small-headed clover	<i>Trifolium microcephalum</i>	2026-05-15
Superb Mariposa Lily	<i>Calochortus superbus</i>	2026-06-11
thimble clover	<i>Trifolium microdon</i>	2026-05-15
tomcat clover	<i>Trifolium willdenovii</i>	2026-05-15
Toyon	<i>Heteromeles arbutifolia</i>	2026-06-19
trailing blackberry	<i>Rubus ursinus</i>	2026-06-24
Tree Clover	<i>Trifolium ciliolatum</i>	2026-05-15
valley oak	<i>Quercus lobata</i>	2026-06-24
wavy-leafed soap plant	<i>Chlorogalum pomeridianum</i>	2026-05-20
western sycamore	<i>Platanus racemosa</i>	2026-05-17
white brodiaea	<i>Triteleia hyacinthina</i>	2026-05-15
white-tipped clover	<i>Trifolium variegatum</i>	2026-05-15
Winecup Clarkia	<i>Clarkia purpurea</i>	2026-05-15
Yellow Mariposa Lily	<i>Calochortus luteus</i>	2026-06-11

AI PLANT COMMUNITY SUMMARY - CLAUDE HAIKU

Site Ecology Assessment: 38.3459, -122.3047 This site supports a **mixed oak woodland and grassland community**, dominated by blue oak (**Quercus douglasii**) and California black oak (**Quercus kelloggii**) with understory components of California buckeye, California bay, and diverse native forbs and wildflowers. The species composition—particularly the presence of clay mariposa lily, canyon liveforever, and Phacelia bolanderi—indicates **well-drained upland soils on slopes with seasonal moisture availability**, typical of interior California foothills. Given this assemblage, the site experiences **moderate drainage with seasonal waterlogging unlikely**, though grading activities should avoid concentrated runoff that would degrade these adapted native communities; drainage modifications should be designed to maintain natural moisture patterns rather than accelerate flow. The density of observations (5,051 across 57 species) suggests an ecologically rich site with significant conservation value. **Consult a licensed botanist for site-specific vegetation assessment.**

Source: iNaturalist research-grade observations - Native species filter applied

FIELD HAZARDS - Venomous & Toxic Species within 25mi

635	154	91	1439
Venomous Snakes	Venomous Spiders	Stinging Insects	Toxic Plants
VENOMOUS SNAKES			
SPECIES			OBSERVATIONS
<i>Crotalus oreganus oreganus</i>			635 obs
VENOMOUS SPIDERS			
SPECIES			OBSERVATIONS
<i>Latrodectus hesperus</i>			154 obs
STINGING INSECTS			
SPECIES			OBSERVATIONS
<i>Dasymutilla aureola</i>			91 obs
<i>Dasymutilla californica</i>			91 obs
<i>Dasymutilla sackenii</i>			91 obs
TOXIC PLANTS			

SPECIES	OBSERVATIONS
<i>Toxicodendron diversilobum</i>	1100 obs
<i>Datura wrightii</i>	57 obs
<i>Datura stramonium</i>	57 obs
<i>Datura ferox</i>	57 obs
<i>Conium maculatum</i>	199 obs
<i>Urtica gracilis</i>	83 obs
<i>Urtica gracilis holosericea</i>	83 obs

AI FIELD SAFETY BRIEFING - CLAUDE HAIKU

Field Safety Brief – Northern California Survey Area Your highest-priority hazard is **poison oak** (*Toxicodendron diversilobum*), with nearly 1,500 observations in this region—expect it year-round on disturbed ground and woodland edges. Second priority is the **Northern Pacific rattlesnake** (*Crotalus oreganus oreganus*), most active April through October when they bask; watch for coiled snakes in rocky areas and brush, and wear snake gaiters or high boots. Avoid touching any unfamiliar plants with three leaflets or white berries, wear long sleeves and pants in brushy terrain, and shower immediately after fieldwork to remove plant oils. Black widow spiders (*Latrodectus hesperus*) shelter in dark, undisturbed spaces—check equipment, gloves, and hard hats before use, and be alert in outbuildings or under debris. Always verify current conditions locally before fieldwork.

Source: iNaturalist research-grade observations - Always verify locally before fieldwork

FCC MICROWAVE PATHS — Licensed beam paths crossing within 1320ft of site

3	545ft	3		
Beam Paths	Closest Beam	Fixed P2P		
TX CALL	RX CALL	TYPE	PATH MI	CLOSEST
WBO61	WBO59	Fixed Point-to-Point	—	545ft
KYH37	WQAT627	Fixed Point-to-Point	—	927ft
WRCR329	WQAT627	Fixed Point-to-Point	—	1289ft

Source: FCC ULS microwave license database — PostGIS spatial query — Paths within 1320ft (0.25mi) of site — Towers may be 50+ miles away — FCC Part 101 licensed paths

FAA AIRWAYS — ATS Routes crossing site area

26	13	2	11			
Total Airways	Jet Routes	Victor Airways	Other Routes			
AIRWAY	TYPE	LEVEL	MEA EAST	MEA WEST	TRACK	REV TRACK
J1	Jet Route	High	18,000 ft	—	303.8°	121.4°
J110	Jet Route	High	18,000 ft	—	137.9°	318.2°
J126	Jet Route	High	18,000 ft	—	325.5°	144.0°
J143	Jet Route	High	18,000 ft	—	325.0°	145.8°
J3	Jet Route	High	18,000 ft	—	342.8°	161.8°
J32	Jet Route	High	18,000 ft	—	19.2°	199.6°
J501	Jet Route	High	18,000 ft	—	317.1°	135.4°
J58	Jet Route	High	18,000 ft	—	65.3°	245.9°
J65	Jet Route	High	18,000 ft	—	304.1°	121.1°
J80	Jet Route	High	18,000 ft	—	65.3°	245.9°
J84	Jet Route	High	18,000 ft	—	50.8°	231.0°
J88	Jet Route	High	18,000 ft	—	308.0°	127.3°
J94	Jet Route	High	18,000 ft	—	65.3°	245.9°
Q1	Q-Route	High	—	—	334.5°	154.1°
Q120	Q-Route	High	—	—	38.1°	219.8°
Q122	Q-Route	High	—	—	37.8°	219.6°
Q124	Q-Route	High	—	—	37.8°	219.6°
Q138	Q-Route	High	—	—	47.7°	229.0°
Q3	Q-Route	High	—	—	164.0°	344.6°
Q5	Q-Route	High	—	—	161.4°	341.7°
T257	T-Route	Low	—	—	321.1°	140.5°
T259	T-Route	Low	—	—	41.6°	221.8°
T261	T-Route	Low	—	—	31.4°	211.7°
T263	T-Route	Low	—	—	308.8°	128.2°
V107	Victor Airway	Low	7,000 ft	—	295.4°	114.2°

AIRWAY	TYPE	LEVEL	MEA EAST	MEA WEST	TRACK	REV TRACK
V108	Victor Airway	Low	4,500 ft	—	117.7°	296.8°

Source: FAA ATS Route FeatureServer — Airways shown cross the site bounding area and may not pass directly overhead — MEA = Minimum En Route Altitude

RAILROAD PROXIMITY — FRA North American Rail Network

8

Rail Lines

OWNER / OPERATOR	TYPE	TRACKS	PASSENGER	STRACNET	SUBDIVISION
CFNR	Other	1	No	No	—
CFNR	Yard	1	No	No	—
CFNR	Mainline	1	No	No	VALLEJO
SMRT	Mainline	1	No	No	BRAZOS JCT.
NVRR	Mainline	1	No	No	MARTINEZ
USG	Other	1	No	No	—
CFNR	S	1	No	No	—
NVRR	Yard	1	No	No	—

Source: FRA North American Rail Network (NARN) — Updated July 2025 — STRACNET = Strategic Rail Corridor Network (DOD)

WILDLIFE & NATURAL HISTORY - iNaturalist within 25mi

200

Total Observations

89

Total Species

64

Birds

5

Mammals

14

Reptiles

6

Amphibians

BIRDS (64 species)

COMMON NAME	Scientific Name	OBS
Wild Turkey	<i>Meleagris gallopavo</i>	10
Snowy Egret	<i>Egretta thula</i>	7
California Quail	<i>Callipepla californica</i>	5
Western Bluebird	<i>Sialia mexicana</i>	5
California Towhee	<i>Melospiza crissalis</i>	5
Great Egret	<i>Ardea alba</i>	4
Red-tailed Hawk	<i>Buteo jamaicensis</i>	4
Black Phoebe	<i>Sayornis nigricans</i>	4
Northern Flicker	<i>Colaptes auratus</i>	3
Cooper's Hawk	<i>Astur cooperii</i>	3
Ash-throated Flycatcher	<i>Myiarchus cinerascens</i>	3
Western Screech-Owl	<i>Megascops kennicottii</i>	3
Turkey Vulture	<i>Cathartes aura</i>	3
Mallard	<i>Anas platyrhynchos</i>	3
Bushtit	<i>Psaltiriparus minimus</i>	3

+ 49 additional species

MAMMALS (5 species)

COMMON NAME	Scientific Name	OBS
Coyote	<i>Canis latrans</i>	3
Black-tailed Jackrabbit	<i>Lepus californicus</i>	3
Mule Deer	<i>Odocoileus hemionus</i>	2
Eastern Fox Squirrel	<i>Sciurus niger</i>	1
Striped Skunk	<i>Mephitis mephitis</i>	1

REPTILES (14 species)

COMMON NAME	Scientific Name	OBS
Western Fence Lizard	<i>Sceloporus occidentalis</i>	10
Pacific Gopher Snake	<i>Pituophis catenifer catenifer</i>	7
Northern Pacific Rattlesnake	<i>Crotalus oreganus oreganus</i>	6
California King Snake	<i>Lampropeltis californiae</i>	3
Pond Slider	<i>Trachemys scripta</i>	2
Western Skink	<i>Plestiodon skiltonianus</i>	2

COMMON NAME	Scientific Name	OBS
Ring-necked Snake	<i>Diadophis punctatus</i>	1
Common Garter Snake	<i>Thamnophis sirtalis</i>	1
Western Yellow-bellied Racer	<i>Coluber constrictor mormon</i>	1
Pacific Ringneck Snake	<i>Diadophis punctatus amabilis</i>	1
Northwestern Pond Turtle	<i>Actinemys marmorata</i>	1
Common Sharp-tailed Snake	<i>Contia tenuis</i>	1
North American Racer	<i>Coluber constrictor</i>	1
Southern Alligator Lizard	<i>Elgaria multicarinata</i>	1

AMPHIBIANS (6 species)

COMMON NAME	Scientific Name	OBS
Pacific chorus frog	<i>Pseudacris regilla</i>	5
American Bullfrog	<i>Lithobates catesbeianus</i>	3
Foothill Yellow-legged Frog	<i>Rana boylei</i>	3
Western Toad	<i>Anaxyrus boreas</i>	2
Rough-skinned Newt	<i>Taricha granulosa</i>	1
California Giant Salamander	<i>Dicamptodon ensatus</i>	1

AI WILDLIFE & ECOLOGICAL ASSESSMENT - CLAUDE HAIKU

Ecological Character Assessment – Site at 38.3459, -122.3047 This site in Sonoma County, California supports a diverse wildlife community characteristic of mixed oak woodland and grassland habitat with nearby aquatic features, as evidenced by the presence of both upland species (Wild Turkey, California Quail, Mule Deer) and riparian/wetland-dependent species (egrets, pond turtles, chorus frogs). Several species present warrant regulatory attention: the Foothill Yellow-legged Frog and California Giant Salamander are California Species of Special Concern with sensitivity to water quality and streamside conditions; the Northwestern Pond Turtle is similarly stream/pond-dependent; and the presence of Northern Pacific Rattlesnake should inform site safety protocols. The reptile and amphibian diversity, particularly aquatic species, indicates proximity to perennial or seasonal water resources (streams, ponds, or wetlands), while the bird and mammal assemblage reflects a mosaic of woodland, scrub, and grassland in moderate condition. The moderate observation density and breadth of species richness suggest relatively intact habitat connectivity with acceptable disturbance levels, though presence of invasive American Bullfrog and Eastern Fox Squirrel indicates some baseline habitat alteration. Development review should prioritize protection of riparian corridors, aquatic breeding habitat, and intact grassland-woodland ecotones to maintain this community structure. **For planning purposes only. Consult a qualified biologist for site-specific biological assessments.**

Source: iNaturalist research-grade observations — Data reflects community science observations and may not represent complete species inventory

✳️ PASSIVE + ACTIVE SOLAR PLANNING — Lat 38.35°

1 • AVERAGE SUN HOURS

Period	Peak Hrs/Day	Notes
Annual average	6.6 hrs	Overall solar resource
Summer average	9 hrs	June–August
Winter average	3.9 hrs	December–February
Worst month (Dec)	3.2 hrs	Size battery storage to this

2 • IDEAL SOLAR ARRAY PITCH

Optimal tilt = 38.3° (latitude) → 9/12 pitch

Pitch	Angle	% of Optimal	Notes
4/12	18.4°	97%	good
5/12	22.6°	97.6%	good
6/12	26.6°	98.2%	↪ excellent
7/12	30.3°	98.8%	↪ excellent
8/12	33.7°	99.3%	↪ excellent
9/12	36.9°	99.8%	↪ annual optimal
10/12	39.8°	99.8%	↪ excellent
12/12	45°	99%	↪ excellent

Going from 5/12 to 6/12 at framing costs ~\$800–1,200 one time. On a 10kW system that's +723 kWh/yr (~\$108 value/yr). Over 25yr panel life: ~\$2710 total gain.

3 • SOLAR HEAT GAIN — BTU/HR PER SQ FT

Use these values as inputs for eQUEST, EnergyPlus, Manual J, or HERS rating

Surface	Winter	Summer	Design Note
South glass (SHGC 0.4)	51	31	Minimize south glass
South glass (SHGC 0.6)	76	46	Clear/uncoated glass
North glass (any)	10	13	Negligible — diffuse only
East / West glass	27	104	△ Minimize — afternoon load
Roof — dark (absorb 0.95)	236	301	Highest load surface
Roof — light (absorb 0.35)	87	111	Cool roof — significant reduction
Roof — metal (absorb 0.25)	62	79	Best roof option for heat
South wall — dark	94	69	
South wall — light	31	23	Light color = major savings
Hardscape — asphalt	181	301	△ Heat island — avoid near structure
Hardscape — concrete	124	206	Also reflects onto adjacent walls
Hardscape — light pave	67	111	Recommended within 50ft
Pool surface	23	38	Low absorb + evaporative cooling

Concrete reflection onto south wall: ~4 BTU/hr-ft² additional load per 100sf of adjacent hardscape.
Pool specular reflection: angle-dependent — east/west pools can cause morning/afternoon glare into glass.

4 • ROOF PITCH SOLAR LOAD — BTU/HR PER 100 SQ FT (SUMMER)

Pitch	Dark Roof	Light Roof	Reduction vs Flat
0/12	301 BTU	111 BTU	baseline (worst)
4/12	241 BTU	89 BTU	-20% dark / -20% light
6/12	196 BTU	72 BTU	-35% dark / -35% light
8/12	166 BTU	61 BTU	-45% dark / -45% light
12/12	135 BTU	50 BTU	-55% dark / -55% light

Scale: multiply BTU × (your roof sf ÷ 100). Pitch + light color = biggest single cooling move.

5 • THERMAL TIME LAG — WALL ASSEMBLY

Peak outdoor temp ~15:00. Heat arrives at interior = peak + lag hours.

Assembly	Lag	Peak Interior Load	Note
2x4 frame (R-15)	0.6 hrs	~15:36 PM	R-value only — no mass benefit
2x6 frame (R-23)	0.9 hrs	~15:54 PM	R-value only — no mass benefit
2x8 frame (R-30)	1.2 hrs	~16:12 PM	R-value only — no mass benefit
2x10 frame (R-37)	1.5 hrs	~16:30 PM	good
2x12 frame (R-44)	1.8 hrs	~16:48 PM	good
ICF 6" core	7 hrs	~22:00 PM	✓ shifts past sunset — natural vent window
8" CMU block	6 hrs	~21:00 PM	✓ shifts past sunset — natural vent window
12" concrete	8.5 hrs	~23:30 PM	✓ shifts past sunset — natural vent window

In 3C Marine: ICF or CMU shifts peak load past sunset. Open windows, flush heat naturally. AC load drops significantly.

6 • TREE SHADOW PLANNING — TREE DUE SOUTH, 100FT FROM STRUCTURE

Tree Ht	Shadow Reaches	Summer hrs/day	Equinox hrs/day	Winter hrs/day	WUI note
10ft	All year	1.1	1	1.2	OK
20ft	All year	2.1	1.9	2.4	OK
30ft	All year	3.1	2.9	3.7	Zone 2 check
40ft	All year	4	3.8	5.2	△ verify
50ft	All year	4.8	4.6	7.3	△ verify
60ft	All year	5.6	5.5	9.3	✗ Zone 1

Deciduous tree: winter solar penetration ~85% (leaves off) — best of both worlds in hot-dry climates.
Evergreen tree: winter penetration ~15% — good windbreak, poor solar strategy south of structure.
△ WUI zones: tree placement must be coordinated with defensible space requirements. Verify with local fire authority — WUI boundaries vary by jurisdiction.

7 • ATTIC VENTING — PER 100 SQ FT OF ROOF AREA

Standard	Unbalanced (1/150)	Balanced† (1/300)	Notes
Code minimum	8 sq in	4 sq in	Moisture protection only
Thermal comfort	24 sq in	12 sq in	Keeps attic within 20°F of outdoor
WUI mesh 1/16"	60 sq in	30 sq in	2.5× compensation — mesh clogs
WUI intumescent	24 sq in	12 sq in	Self-closing — no compensation needed

Pitch factor: 0/12×1.0 · 4/12×0.80 · 6/12×0.65 · 8/12×0.55 · 12/12×0.45
Color factor: Dark×1.0 · Medium×0.75 · Light×0.55 · Metal×0.45
Scale: multiply sq in × (your roof sf ÷ 100) × pitch factor × color factor
†**Balanced** = ≥40% NFA within top third of roof height (ridge) + ≥40% within bottom third (soffit)
WUI conflict: Soffit vents are ember collection points. In high WUI zones — eliminate soffit vents, use ridge-only (unbalanced/1/150). In low WUI — balanced system preferred.
△ Verify WUI zone with local fire authority — boundaries vary significantly by jurisdiction.

🔗 AI SOLAR & PASSIVE DESIGN SUMMARY — HOMEOWNER GUIDE

Your Home's Solar & Energy Plan Your location gets about 6.6 hours of strong sun per day on average—think of it as having a reliable solar "battery charge window" that's long enough to make panels worthwhile, especially in summer when you hit nearly 10 hours. Your roof's pitch is already nearly perfect for solar at 38 degrees, which is lucky. Now here's what surprises most homeowners: that dark roof absorbs about 300 BTUs of heat per hour per square foot in summer, but if you went light-colored, you'd cut that almost in half—so a simple re-roofing choice can rival air conditioning in its impact on your cooling costs. Your west-facing windows are the sneaky culprit nobody wants to hear about; they're like an open oven door facing your living room in late afternoon, dumping 100+ BTUs of afternoon heat right when it's hottest outside. The good news is that a 40-foot deciduous tree on your south side would naturally shade your house for about 4 hours in summer (blocking that excess heat), then drop its leaves in winter to let the gentle 51 BTU solar gain through your south windows help warm the house—it's free seasonal climate control. Finally, thermal mass is simply your home's ability to soak up heat during the day and release it slowly at night, like a sponge; thicker, denser walls (like ICF concrete) do this much better than lightweight framing, which matters because your mild marine climate has smaller temperature swings to manage. These numbers are planning estimates—your architect or energy modeler can plug them directly into their software for a full analysis.

Planning estimates only — clear-sky model, standard assumptions. Values formatted for eQUEST / EnergyPlus / Manual J input. Verify with licensed architect, energy modeler, or solar installer before specifying.

Technical Brief — for architects, engineers, and surveyors

SITE INTELLIGENCE BRIEFING **38.34586°N, 122.30466°W | Napa County, California**

Structural Design Drivers & Climate Envelope

This site falls within IECC Climate Zone 3C (Marine) with an Annual Freezing Index of 0°F-days, indicating negligible freeze-thaw cycling and no frost-related foundation uplift risk. Wind and snow data are not available for this location; however, the marine classification and zero AFI suggest a temperate coastal or bay-adjacent microclimate where wind loads may be moderate and snow loading is minimal to absent. Without quantified wind speed or snow load data, the design team cannot yet determine whether wind or seismic response governs the lateral system; this gap should be closed via ASCE 7 site-specific wind speed determination and local building department confirmation of design wind velocity. Absent snow and frost depth requirements, the structural envelope is likely constrained by code-minimum dead and live loads and wind/seismic interaction rather than environmental extremes.

Geotechnical Profile & Foundation Implications

The site is underlain by Coombs gravelly loam on 2 to 5 percent slopes, classified as Hydrologic Group C with well-drained status. Group C soils typically exhibit moderate infiltration rates and moderate permeability; well-drained condition confirms surface water will shed readily and subsurface drainage conditions favor shallow foundations. Soil texture is not specified in available data and should be obtained via Phase I site investigation or USDA NRCS detailed soil series report to confirm bearing capacity, settlement behavior, and expansion potential. The absence of a reported frost depth (AFI = 0°F-days) eliminates the need for frost-protected shallow foundations or deep embedment; conventional spread footings can be designed for seasonal moisture variation and the actual bearing capacity determined by soil testing. Slope gradient of 2 to 5 percent is gentle; stability should be confirmed through site-specific geotechnical evaluation, particularly if grading or cut-fill operations are planned.

Seismic & Subsidence Hazards

No Quaternary faults are recorded within the search radius, and no subsidence basin data exists for the site. Napa County is seismically active regionally (San Andreas system ~30 miles to the east); seismic design remains mandatory per California Building Code, but absence of nearby mapped faults reduces localized hazard from surface fault rupture or displacement. Subsidence is not a documented risk. Wildfire data are not provided. The design team should confirm current seismic design category via USGS hazard maps and IBC/CBC procedures; this will drive lateral system selection and detailing once wind data are also available.

Mining & Industrial Context

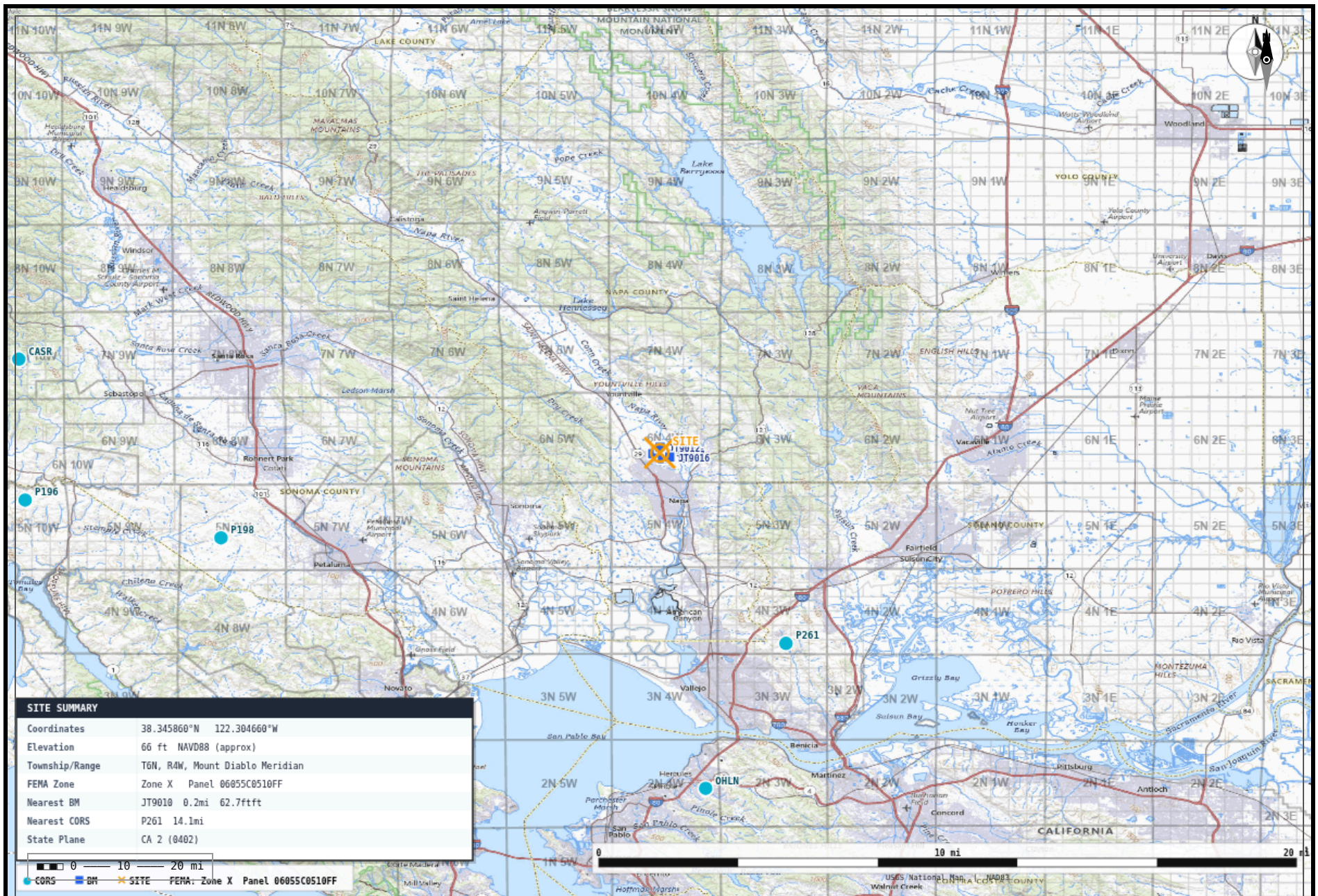
Three active mineral producers operate within 5 miles: Basalt Rock Co (2 miles, clay), Mt George Pumice Deposit (3.1 miles, pumice), and Silva Pumice Quarry (5 miles, pumice). An additional 200 MRDS deposits are recorded within ~35 miles. These operations warrant vibration and dust impact assessment if sensitive uses (residential, laboratory, health care) are planned. Eight oil and gas wells exist within 10 miles; the nearest is a plugged Mobil Oil well 7.2 miles distant. No active oil/gas production is recorded within the search zone, eliminating immediate subsurface pressure or induced seismicity risk. Ground stability beneath active quarries should be researched if site access or future expansion may occur near extraction areas.

Bottom Line

Obtain soil boring data (bearing capacity, expansion index, actual texture) and confirm design wind speed and seismic design category via ASCE 7 and CBC procedures before finalizing the lateral and foundation systems.

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Planning aid only — not a substitute for licensed professional review, geotechnical investigation, or code compliance verification.

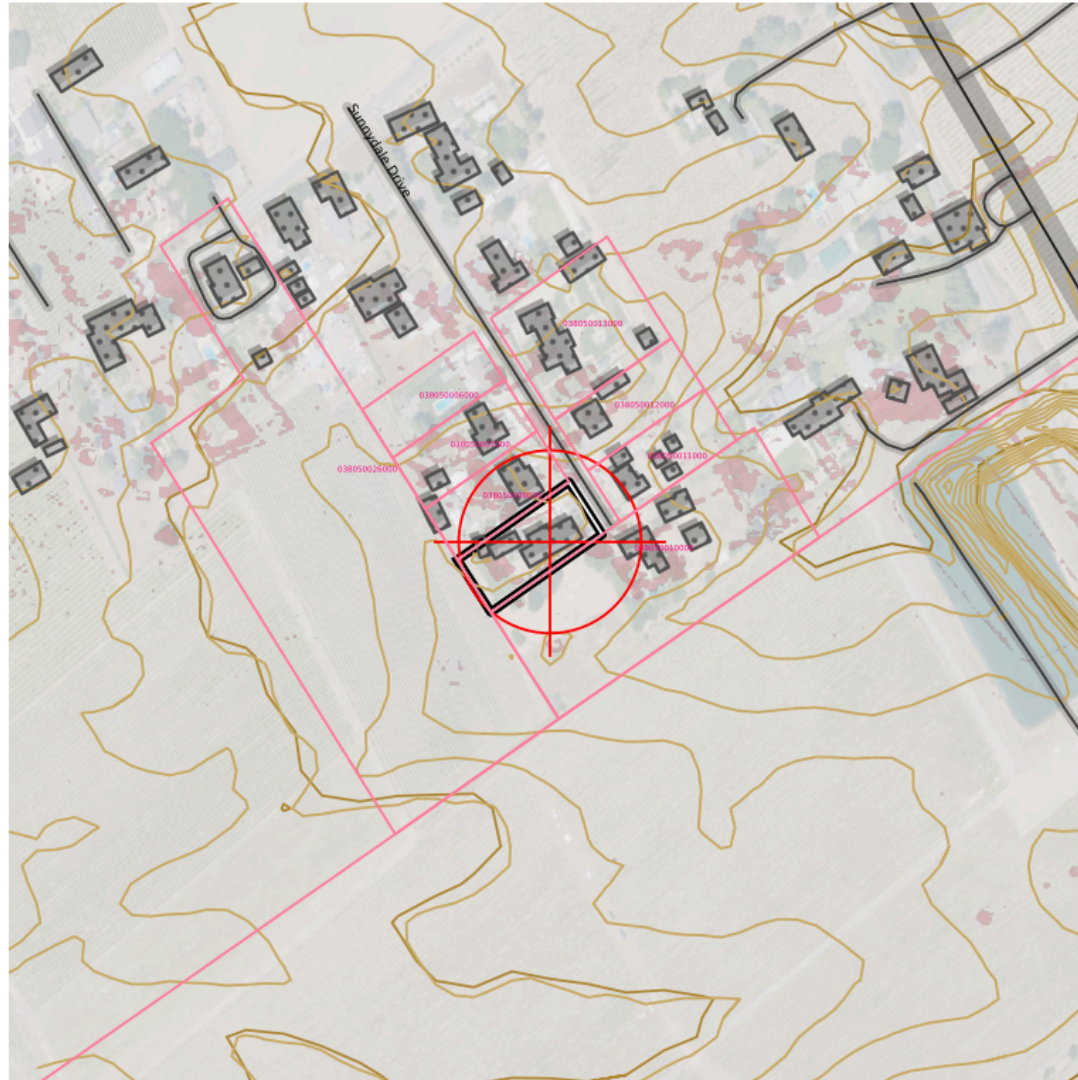


TOPOSCOUT
FIELD MAP

TopoScout Field Map
Lat: 38.34586 Lng: -122.30466
38.345860°N 122.304660°W | Radius: 30 mi

Generated: July 13, 2026
Base: USGS National Map
Generated by TopoScout - outkastdesigns.com

TOPOSCOUT SITE PLAN PREVIEW | \$19 UPGRADE FOR AUTOCAD DXF FILE



INCLUDED LAYERS



0 — 400 ft State Plane · NAD83 · US Feet

- Contours (Index/Major/Minor/1ft)
- PLSS Section/Township/QQ
- Roads Major/Local/Trail + ROW
- Buildings
- Hydro
- Power Lines
- Pipeline
- Railroad
- Benchmarks + Labels
- County Boundary
- OTLS Grid
- Site Info
- Elevation Labels
- 3D Polylines (all geometry)



SITE AERIAL – 600 ft × 600 ft

38.34586°N, 122.30466°W · Google Aerial



AREA CONTEXT — 3000 ft × 3000 ft

38.34586°N, 122.30466°W · Google Aerial

✂ FIELD RESOURCE LINKS

PRE-MISSION PLANNING

Trimble GNSS Planning Online –

<https://trimblegnssplanningonline.com/>

Satellite visibility, DOP charts, sky plots

FAA B4UFLY –

https://www.faa.gov/uas/recreational_flyers/where_can_i_fly/b4ufly

Drone airspace authorization, TFRs, restricted areas

Aloft (Kittyhawk) – <https://www.aloft.ai/>

Field-friendly drone airspace map and LAANC authorization

NOAA Space Weather — Live Kp –

<https://www.swpc.noaa.gov/products/planetary-k-index>

Real-time Kp index, geomagnetic storm alerts

POST-PROCESSING

NGS OPUS – <https://geodesy.noaa.gov/OPUS/index.jsp>

Online Positioning User Service – static GNSS processing

NGS OPUS-RS – <https://geodesy.noaa.gov/OPUSI/index.jsp>

Rapid static processing – sessions as short as 15 minutes

CORS Data Download – <https://geodesy.noaa.gov/CORS/cors-data.shtml>

Raw RINEX observation files from CORS stations

FEMA & FLOOD

FEMA Flood Map Service Center –

<https://msc.fema.gov/portal/home>

FIRM panels, flood zone lookup by address or coordinates

FEMA Elevation Certificate – [https://www.fema.gov/flood-](https://www.fema.gov/flood-insurance/work-with-nfip/elevation-certificate)

[insurance/work-with-nfip/elevation-certificate](https://www.fema.gov/flood-insurance/work-with-nfip/elevation-certificate)

Current EC form, instructions, LOMA/LOMR submission guidance

BASE STATION & NETWORK

NOAA CORS Network – <https://geodesy.noaa.gov/CORS/cors-data.shtml>

Continuously operating reference stations, data download

NGS CORS Station Map –

<https://geodesy.noaa.gov/CORS/GoogleMap/CORSmap.shtml>

Interactive map of all CORS stations with status

IGS Station Monitor – <https://www.igs.org/network>

International GNSS Service – global base station health

REFERENCE & CALCULATIONS

NGS Datasheets – <https://www.ngs.noaa.gov/cgi-bin/datasheet.prl>

Benchmarks, control points, passive monuments

NGS Coordinate Conversion (NCAT) –

<https://geodesy.noaa.gov/NCAT/>

NADCON5, datum transforms, state plane conversions

NOAA Magnetic Declination –

<https://www.ngdc.noaa.gov/geomag/calculators/magcalc.shtml>

Declination calculator for any location and date

NGS Geoid Models – [https://geodesy.noaa.gov/GEOID/](https://geodesy.noaa.gov/GEOID/GEOID18)

[GEOID18](https://geodesy.noaa.gov/GEOID/GEOID18) and current model downloads, GEOID calculator

DRONE OPERATIONS

FAA Part 107 — Commercial Rules –

https://www.faa.gov/uas/commercial_operators

Full Part 107 regulatory summary for commercial UAS operations

FAA DroneZone – <https://faadronezone.faa.gov/>

Drone registration, operator certificates, waiver management

FAA B4UFLY –

https://www.faa.gov/uas/recreational_flyers/where_can_i_fly/b4ufly

Pre-flight airspace check, TFRs, LAANC authorization

FAA Part 107 Waivers –

https://www.faa.gov/uas/commercial_operators/part_107_waivers

Waiver applications for operations beyond standard 107 rules

Aloft — Field Airspace Map – <https://www.aloft.ai/>

Mobile-friendly real-time airspace, LAANC, flight logging

AIR QUALITY & WILDFIRE SMOKE

[AirNow Fire & Smoke Map](https://www.fire.airnow.gov)

<https://www.fire.airnow.gov>

Current fire locations, smoke plumes, PM2.5 readings by location

[AirNow Interactive Map](https://gispub.epa.gov/airnow)

<https://gispub.epa.gov/airnow>

Real-time AQI monitors, hourly NowCast, 48-hr forecast overlays

[NOAA Smoke Forecast](https://airquality.weather.gov)

<https://airquality.weather.gov>

48-hr smoke concentration forecast, HRRR-Smoke model

